



PHYSICAL AI: The Missing Layer In Robotics

What happens when AI leaves the screen and enters the real world? From humanoid robots to autonomous machines, physical AI is turning intelligence into action.



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The simplest way to explain physical AI is as a combination of software (the mind) and hardware (the body). In this case, the hardware is a robot, and the software is a form of artificial intelligence (AI). When a robot and AI are combined, the result is what is known as Embodied AI, which enables intelligence to operate in a physical form and interact with the world.

Most AI systems focus on digital environments, meaning they can process and manipulate information, including text, images, videos, and large volumes of data. They help people by providing information, automating processes, and assisting with decision-making.

On the other hand, physical AI is more of a combination of robotics and artificial intelligence; it can also be developed and tested in digital environments known as simulations. These simulations allow robots to be trained and evaluated safely and efficiently before deployment in the real world.

Why now?

With the emergence of foundation models, the ability to perform physical tasks

became significantly more flexible, similar to how large language models (LLMs) such as ChatGPT and Gemini expanded the capabilities of language-based AI.

Within a few years, this adaptable intelligence had begun making AI physical in practical ways. Today, vision-language models (VLMs), vision-language-action (VLA) models, and world models enable robots to perceive their surroundings, understand instructions, and act autonomously in real-world environments.

The emergence of adaptable intelligence has made physical AI a significant new frontier. Until recently, many robotics companies remained in research and prototyping phases for nearly two decades, as the technology required for widespread commercial deployment and large-scale adaptability simply did not exist.

AI has effectively become the missing layer that transforms robotics from rigid automation into scalable, intelligent systems.

The focus has also shifted from experimentation to utility. End users care less about the sophistication of underlying models and more about whether systems solve real-world problems efficiently and